

## 2.1 Introduction to Optimization

Cost functions, local versus global minima, convexity, and first- and second-order tests.

### 1. The problem

Pick a set ( $S$ ) (usually  $\mathbb{R}^l$ ) and a cost ( $J: S \rightarrow \mathbb{R}$ ). Solve

$$J(\hat{\theta}) = \min_{\theta \in S} J(\theta).$$

Maximizing profit ( $P$ ) is minimizing ( $-P$ ). This course always writes a **min**.

Linear regression is the first example. Model ( $y = \theta^T \mathbf{x} + \epsilon$ ).

Average squared error

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \bigl( \theta^T \mathbf{x}^{(i)} - y^{(i)} \bigr)^2.$$

Each gradient step is “make the residuals smaller.”

### 2. Local versus global

A **local** minimum is lowest in a neighborhood. A **global** minimum is lowest on all of ( $S$ ).

On a mountain trail, a valley is local. Sea level in the whole range is global.

If ( $J$ ) is **convex**,

$$J(t\theta + (1-t)\beta) \leq tJ(\theta) + (1-t)J(\beta),$$

every local min is global. That is the property that makes Module 1 feel safe.

| Convex costs (this course)     | Not convex     |
|--------------------------------|----------------|
| Linear regression (MSE)        | Deep nets      |
| Logistic regression (log loss) | (k)-means      |
| Soft-margin SVM                | GMM likelihood |

### 3. First- and second-order tests

**First-order (necessary).** At a local min of a smooth unconstrained ( $J$ ),  $\nabla J(\hat{\theta}) = \mathbf{0}$ . The slope is flat.

**Second-order (sufficient).** If also  $\nabla^2 J(\hat{\theta})$  is positive definite, you have a local min.

A zero gradient is not enough: a saddle is flat and is not a min. The Hessian tells the two cases apart. Week 2's algorithms either follow  $(-\nabla J)$  or invert  $(\nabla^2 J)$ .